

Truth Learning in a Social Network

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Acknowledgements

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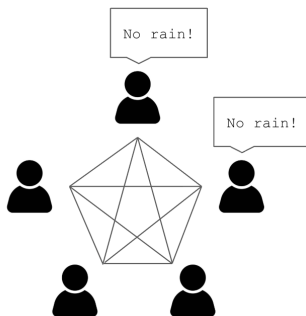
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Background

- In social settings, we often make decisions based on what we see others do, combined with our own private beliefs

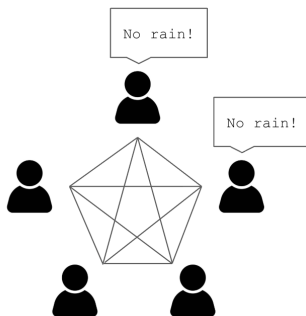
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- For general networks, the strength of your decision may depend on the network structure & decision ordering

- Networks are infinite families $\mathcal{F} = \{G_n, \sigma_n\}_n$ of graphs G_n and orderings σ_n . Agents (vertices) can only see their neighbor's decisions
- Ground truth $\Theta \in \{0, 1\}$ is an unknown random variable
- Agents are given independent private signals $p_v \in \{0, 1\}$ each correlated with Θ with known parameter $q > 1/2$
- Each agent broadcasts their prediction in an order according to σ
- Agents are perfectly Bayesian & predict the most probable truth given their knowledge of G , σ , q , and their neighbors' signals
- The learning rate $\ell(v)$ of a vertex under an ordering σ is the probability that it correctly predicts the ground truth. The learning rate $L(G, \sigma)$ is the average learning rate of all vertices.

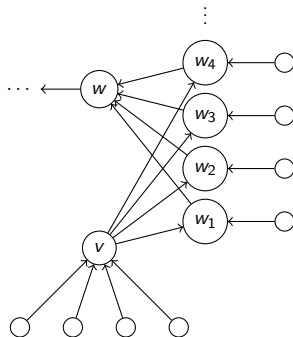
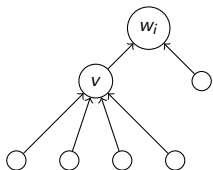
Asymptotic truth learning: $L(G_n, \sigma_n)$ approaches 1 as $n \rightarrow \infty$.

Research Questions

However, networks with a fixed ordering can be surprisingly fragile!

Claim

There exists a network $\mathcal{F}$ that learns iff each agent's private signal strength is $q < q_0$, for some threshold $q_0 \approx 0.79$.



Instead, random orderings seem to be much more “robust”.

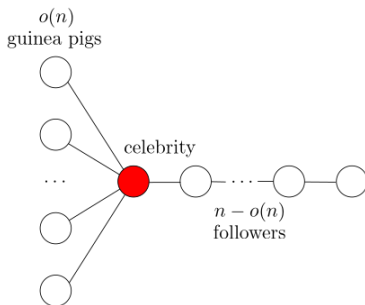
- The random learning rate $L(G)$ of a graph G with n vertices is the average learning rate over all $n!$ orderings.
- A network achieves *random* learning if $L(G_n) \rightarrow 1$ as $n \rightarrow \infty$.

Central Question

What are the necessary/sufficient conditions for a social network to support asymptotic truth learning under random orderings?

Question #1

Given a network obtains truth learning under random orderings, how many modifications can it sustain without losing its learning properties?



Adversarial Robustness

A random-order learning network with learning rate $1 - \varepsilon$ is robust to $o(1/\varepsilon)$ many modifications (edge/vertex deletion/insertions, adversaries).

Claim (monotonicity)

Conditioned on being at index $i \in [n - 1]$ in the random decision ordering, the learning rate of any vertex v is at most that of its learning rate conditioned on being at index $i + 1$.

Theorem

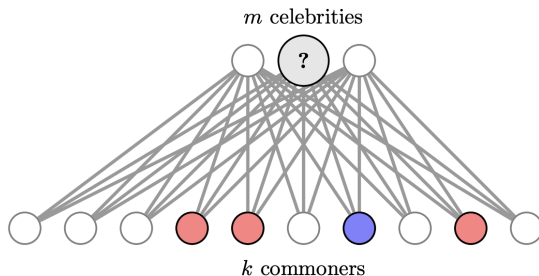
Given a graph G with random-order learning rate $1 - \varepsilon$, after k modifications, G obtains learning rate $1 - O(k\varepsilon)$

Regarding tightness, we note that there exist random-order learning networks that fail to learn after $\Omega(1/\varepsilon)$ strategic modifications.

Structural Properties of a Learning Network

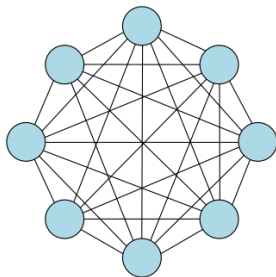
Question #2

Given that a network obtains truth learning, what structural properties must it necessarily have? (e.x. large independent sets, high connectivity)



Structural Properties of a Learning Network

To achieve learning, we need $\omega(1)$ many vertices to announce their private signal, which isn't possible when the max independent set is of size $O(1)$.

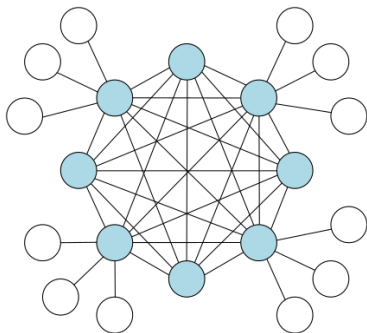


Claim

Given that $\mathcal{F}$ obtains truth learning (either strategic/random-order), the $G_n \in \mathcal{F}$ must have an independent set of size $\omega(1)$.

Structural Properties of a Learning Network

Interestingly, a network may depend on a negligible number of poorly connected, non-learning “guinea pigs”.

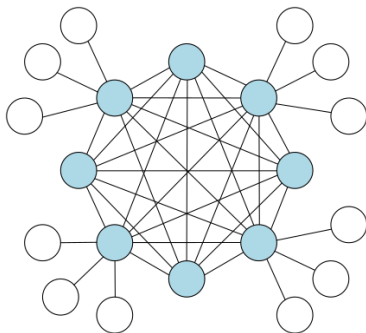


Claim

There exists a network $\mathcal{F}$ obtaining random-order learning that no longer achieves learning when all vertices of degree 1 are removed.

Question #3

For an arbitrary network, how many modifications do we need to make it ensure it achieves random-order learning?



Engineering Random Order Learning Networks

Definition

Let $f(v, t) : V \times (0, 1) \rightarrow \{0, 1\}$ be a "learning oracle" that outputs 1 if v learns at rate at least t under random orderings, and 0 otherwise.

Theorem

Given efficient query access to f , there exists an algorithm running in polynomial time that modifies a graph to obtain random-order learning. Furthermore, the number of modifications needed is a $g(n) \log n$ -approximation of the optimum, for any function $g(n) = \omega(1)$.

Idea: choose specific vertices to 'boost' by connecting them to $\omega(1)$ independent guinea pigs.

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- We want to boost the minimum number of vertices such that the boosted vertices influence almost all nonlearning vertices (on average).
- This is very similar to the set cover problem, which has a $\log n$ greedy approximation!
- This problem also be written as a *submodular* function, so by the same techniques we also have a $\log n$ greedy approximation.
- We pick the next vertex to boost by calculating the expected number of *new* nonlearning vertices it can reach, by sampling random orderings of the vertices.

Question #4

Is determining whether a graph/vertex obtains a learning rate above a certain threshold provably hard?

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